LAPORAN PRAKTIKUM

POSTTEST 7

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

Nathanael Larry Glasia (2509106068)

Kelas (B1‘25)

PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

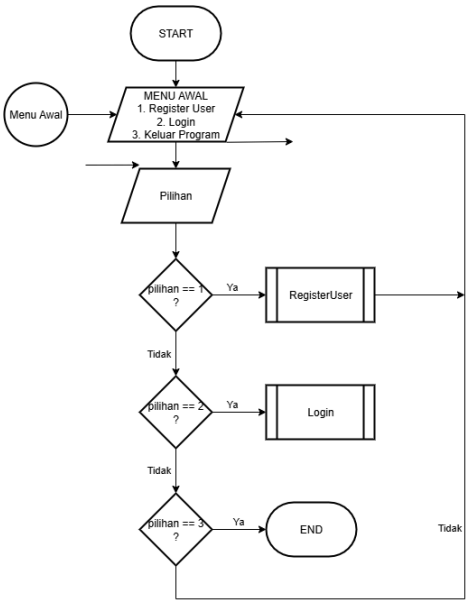
SAMARINDA

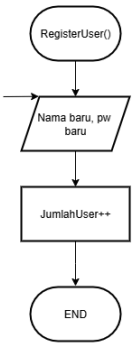
2026

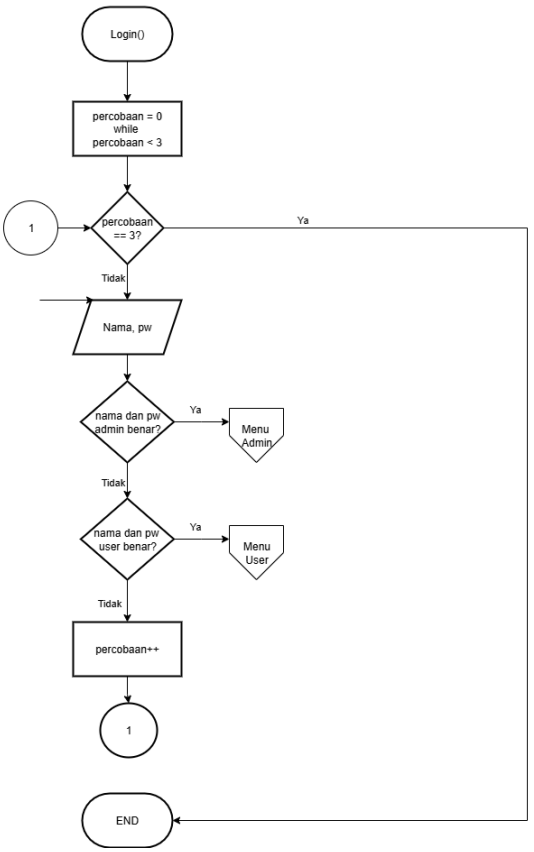
1. Flowchart

Program dimulai. Menu awal memiliki 3 pilihan:

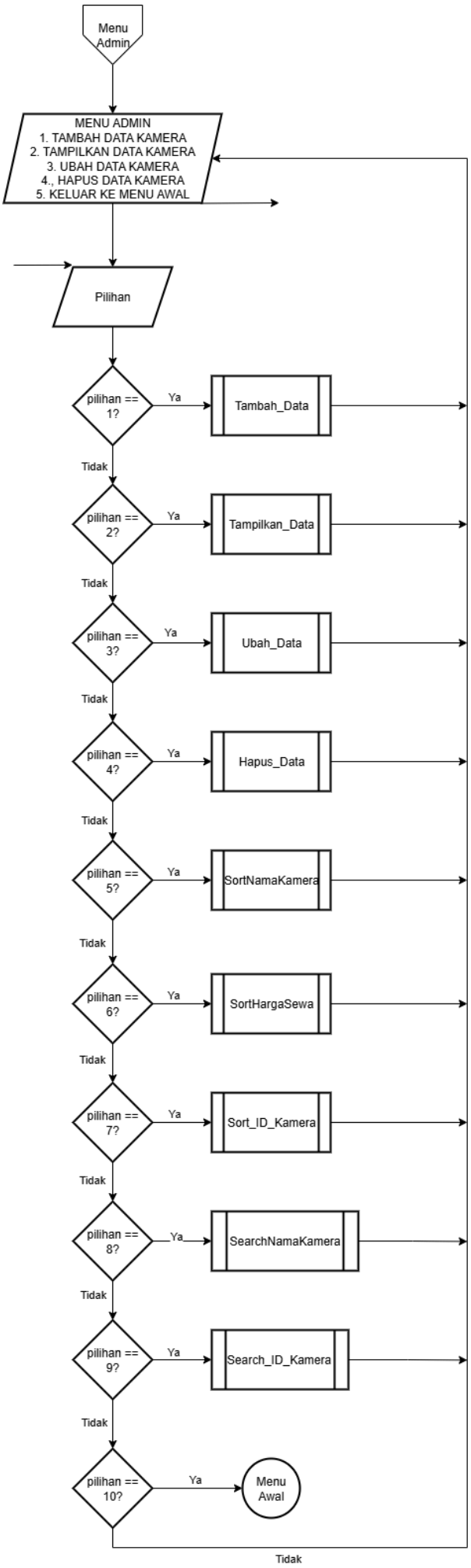
1. Register User
2. Login
3. Keluar Program





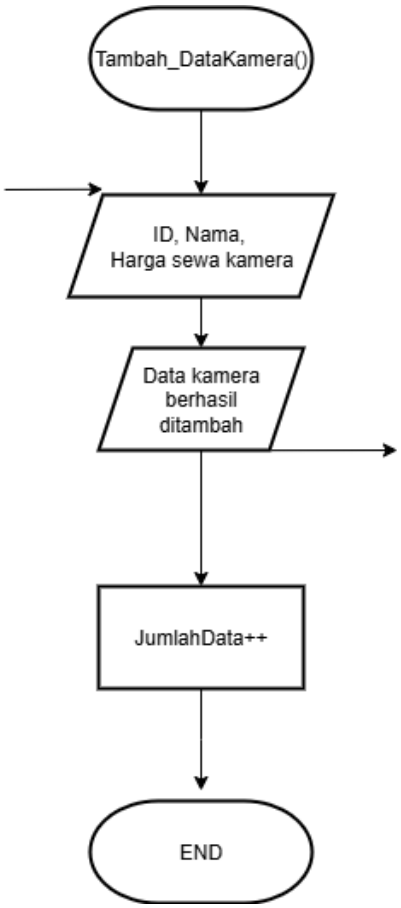


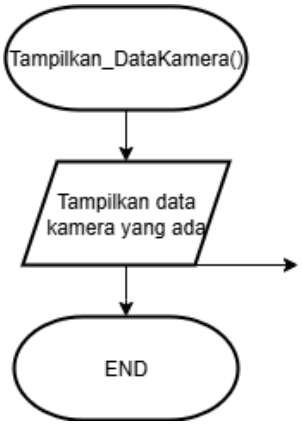
Jika memilih 1, maka pengguna akan diarahkan ke menu register untuk daftar akun baru. Admin diberi nama “LARRY” dan passwordnya “068”, jadi admin tidak perlu register (bisa langsung login ke program). Jika sudah register, pengguna dapat login jika memilih pilihan 2. Di menu login, pengguna juga memasukkan nama dan password yang sesuai ketika pengguna melakukan register akun. Pengguna diberi percobaan login yang diberi nilai 0 dan ada kondisi yang diberikan selama percobaan login kurang dari 3. Jika salah memasukkan nama atau password, maka percobaan login bertambah sebanyak 1 dan jika sampai 3 kali gagal login, maka program otomatis berhenti. Jika memilih 3, maka program akan berhenti.

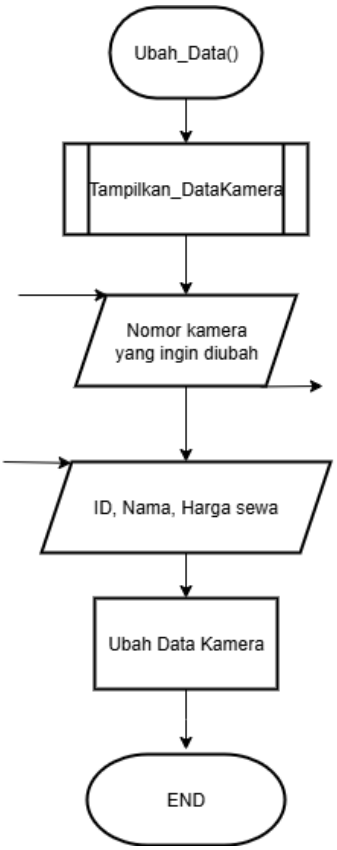

Menu Admin
MENU ADMIN
1. TAMBAH DATA KAMERA
2. TAMPILKAN DATA KAMERA
3. UBAH DATA KAMERA
4., HAPUS DATA KAMERA
5. KELUAR KE MENU AWAL
Pilihan
pilihan == 1?
Ya
Tambah_Data
Tidak
pilihan == 2?
Ya
Tampilkan_Data
Tidak
pilihan == 3?
Ya
Ubah_Data
Tidak
pilihan == 4?
Ya
Hapus_Data
Tidak
pilihan == 5?
Ya
SortNamaKamera
Tidak
pilihan == 6?
Ya
SortHargaSewa
Tidak
pilihan == 7?
Ya
Sort_ID_Kamera
Tidak
pilihan == 8?
Ya
SearchNamaKamera
Tidak
pilihan == 9?
Ya
Search_ID_Kamera
Tidak
pilihan == 10?
Ya
Menu Awal
Tidak

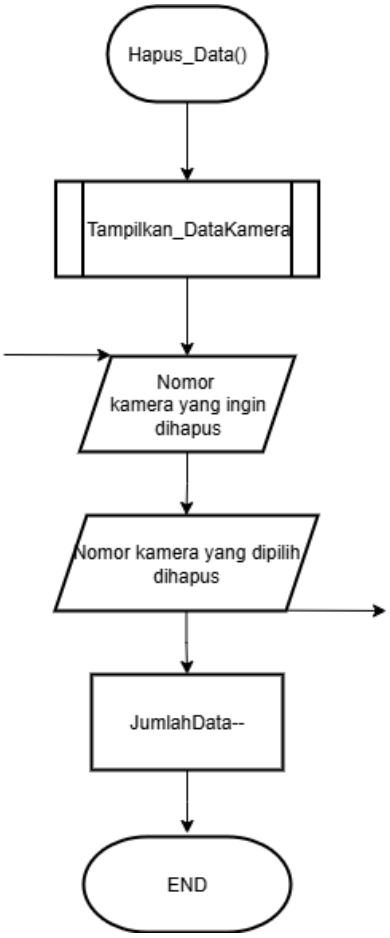
Menu Admin memiliki:

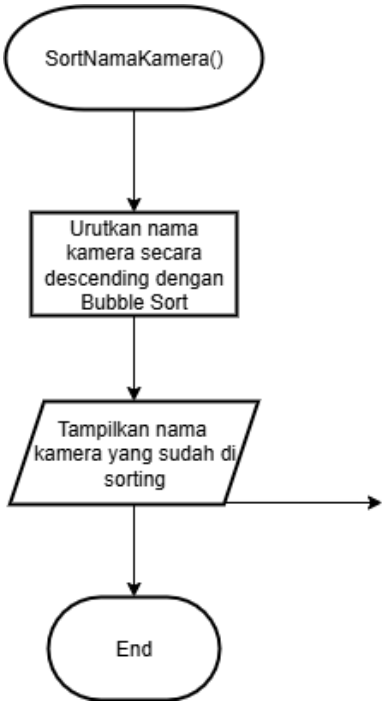
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Sortir Nama Kamera
6. Sortir Harga Sewa Kamera
7. Sortir ID Kamera
8. Search Nama Kamera
9. Search ID Kamera
10. Logout

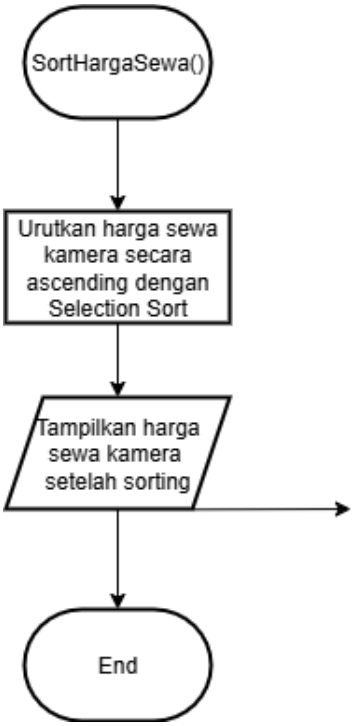


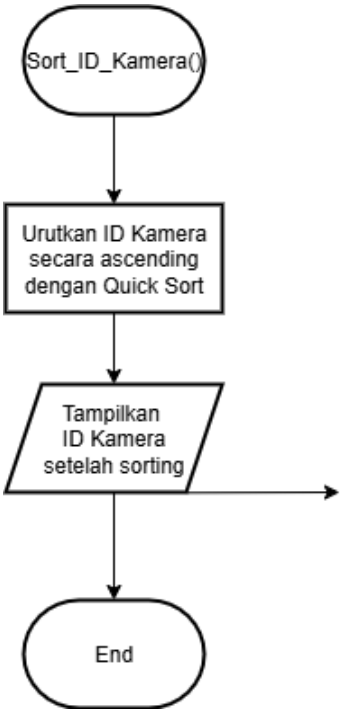

Tampilkan_DataKamera()
Tampilkan data kamera yang ada
END

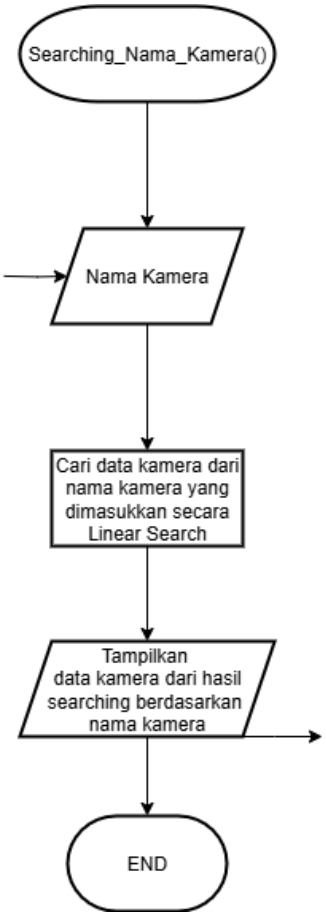

Ubah_Data()
Tampilkan_DataKamera
Nomor kamera yang ingin diubah
ID, Nama, Harga sewa
Ubah Data Kamera
END

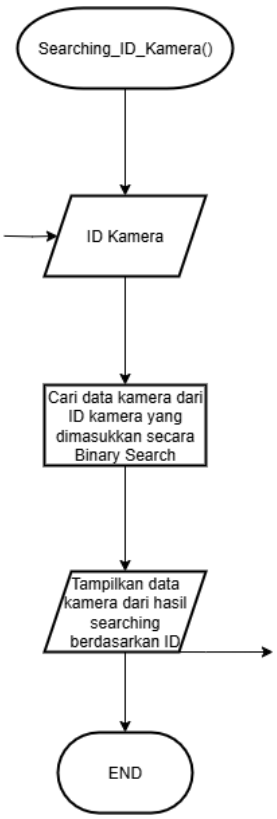

Hapus_Data()
Tampilkan_DataKamera
Nomor kamera yang ingin dihapus
Nomor kamera yang dipilih dihapus
JumlahData--
END


SortNamaKamera()
Urutkan nama kamera secara descending dengan Bubble Sort
Tampilkan nama kamera yang sudah di sorting
End

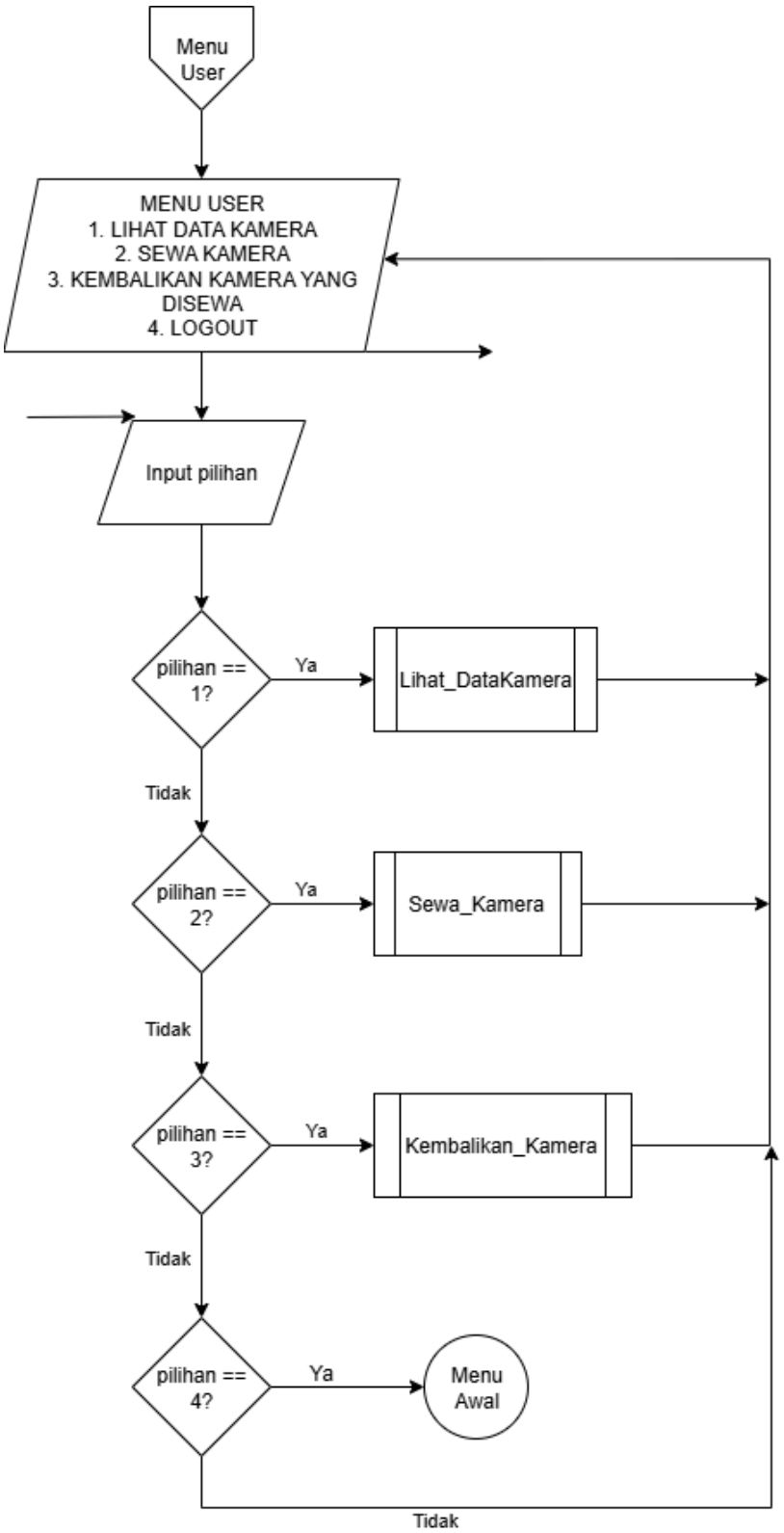

SortHargaSewa()
Urutkan harga sewa kamera secara ascending dengan Selection Sort
Tampilkan harga sewa kamera setelah sorting
End


Sort_ID_Kamera()
Urutkan ID Kamera secara ascending dengan Quick Sort
Tampilkan ID Kamera setelah sorting
End


Searching_Nama_Kamera()
Nama Kamera
Cari data kamera dari nama kamera yang dimasukkan secara Linear Search
Tampilkan data kamera dari hasil searching berdasarkan nama kamera
END

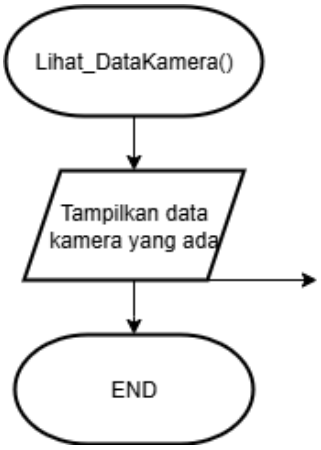


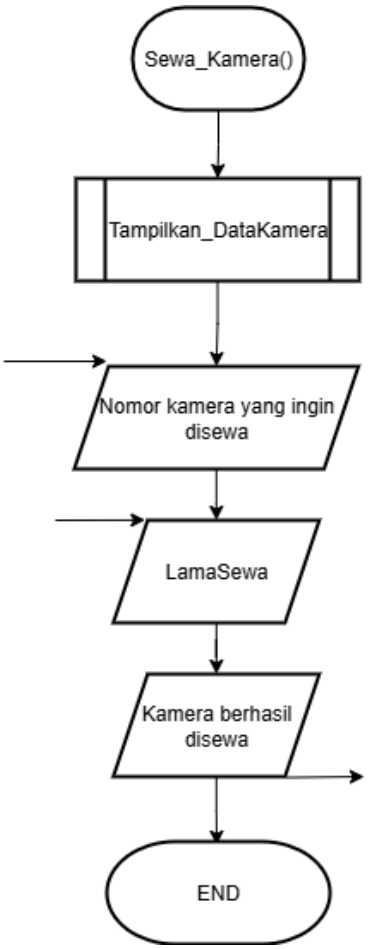
Jika memilih menu nomor 1, maka admin bisa menambah ID kamera, nama kamera, dan harga sewa/hari kamera. Jika memilih menu nomor 2, program akan menampilkan data kamera, baik ID kamera, nama kamera, dan harga sewa per/hari. Jika memilih menu nomor 3, admin dapat mengubah data kamera. Jika memilih menu nomor 4, admin dapat menghapus data kamera. Jika memilih nomor 5, maka admin dapat menyortir nama kamera dengan metode Bubble Sort secara descending. Jika memilih nomor 6, maka admin dapat menyortir harga sewa kamera dengan metode selection sort secara ascending. Jika memilih nomor 7, maka admin dapat menyortir ID Kamera dengan metode Quick Sort secara ascending. Jika memilih nomor 8, maka admin dapat melakukan pencarian berdasarkan nama kamera. Data kamera akan dicari dengan menggunakan metode Linear Search. Jika memilih nomor 9, maka admin dapat melakukan pencarian berdasarkan ID kamera. Data kamera akan dicari dengan menggunakan metode Binary Search. Setelah selesai menggunakan menu 1-9, program akan kembali ke menu admin. Jika memilih menu nomor 10, maka program akan mengarahkan ke menu awal.

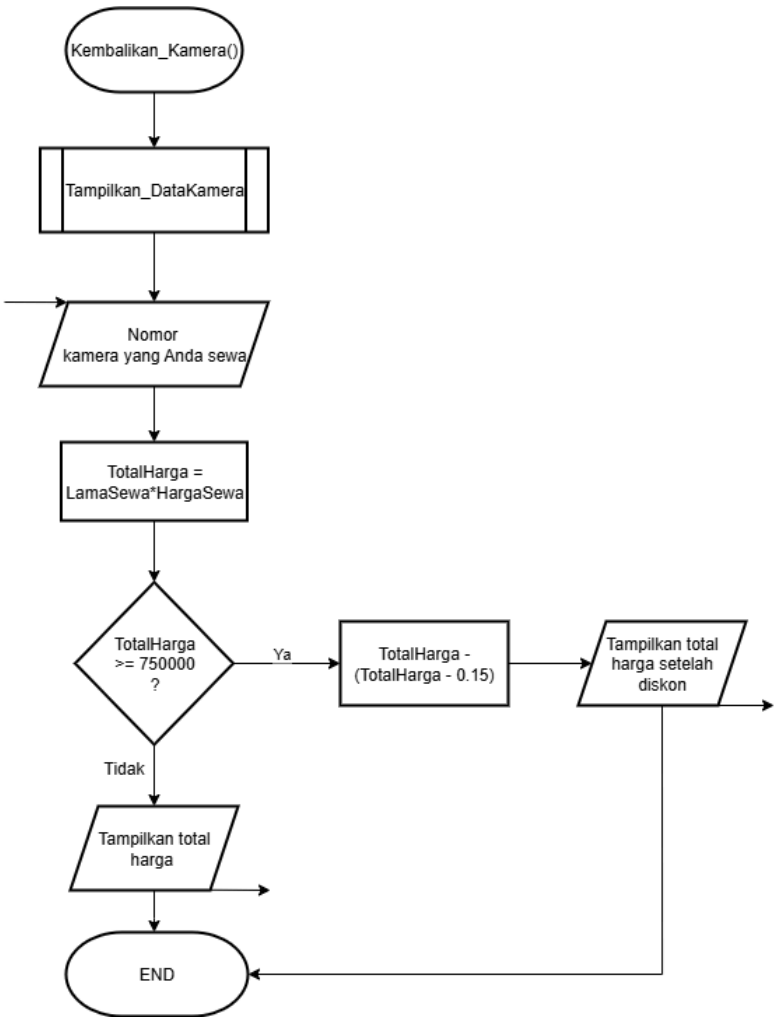

Menu User
MENU USER
1. LIHAT DATA KAMERA
2. SEWA KAMERA
3. KEMBALIKAN KAMERA YANG DISEWA
4. LOGOUT
Input pilihan
pilihan == 1?
Ya
Lihat_DataKamera
Tidak
pilihan == 2?
Ya
Sewa_Kamera
Tidak
pilihan == 3?
Ya
Kembalikan_Kamera
Tidak
pilihan == 4?
Ya
Menu Awal
Tidak

Menu User memiliki:

1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera yang Disewa
4. Logout







Jika memilih nomor 1, user dapat melihat data kamera yang ada. Jika memilih nomor 2, User dapat menyewa kamera yang tersedia. Jika memilih nomor 3, user dapat mengembalikan kamera yang sebelumnya disewa dan program juga akan menghitung total harga sewa berdasarkan harga sewa dan lama sewa. Setelah selesai menggunakan menu 1-3, program akan kembali ke menu user. Jika memilih nomor 4, maka user kembali ke menu awal.

2. Deskripsi Singkat Program

Program Sistem Data Penyewaan Kamera adalah program yang dimana Admin mengelola dan menyimpan data kamera yang terdiri dari ID kamera, nama kamera, harga sewa/hari, dan status kameranya apakah tersedia atau sedang disewa. User dapat menyewa kamera yang sedang tersedia dan dapat mengembalikan kamera sewaannya sesuai dengan kamera yang disewa dan berapa lama User menyewa kamera itu.

Program menggunakan fungsi dan prosedur untuk memecah beberapa menu dari admin dan user. Pada program ini juga terdapat pointer (* dan &). Dereference Operator (*) adalah operator untuk melihat nilai yang tersimpan pada alamat memori. Sedangkan Address-Of Operator adalah operator untuk mendapatkan atau melihat alamat memori yang dimiliki sebuah variabel. Kedua operator juga bisa digunakan sebagai parameter fungsi ataupun nilai kembalian. Pointer pada struct juga diterapkan untuk mengakses serta memanipulasi data yang ada di dalam struct.

Program juga menerapkan sorting untuk mengurutkan data baik secara ascending (terkecil ke terbesar) maupun descending (terbesar ke terkecil). Metode Sort yang digunakan pada program ini adalah Bubble Sort, Selection Sort, dan Insertion Sort. Program ini juga menggunakan metode searching/pencarian untuk mencari data kamera berdasarkan nama kamera atau ID kamera. Metode Searching yang digunakan pada program ini adalah Linear Search dan Binary Search. Pada program ini juga ada error handling untuk memberi informasi pada admin dan user saat melakukan kesalahan pada program sehingga program tetap bisa berjalan dengan baik

3. Source Code

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

struct User
{
    string username;
    string password;
    string role;
};

struct StatusKamera
{
    string kondisi;
};

struct Kamera
{
    int ID_Kamera;
    string NamaKamera;
    int HargaSewa;
    int LamaSewa;
    string NamaPenyewa;
    StatusKamera status;
};

void RegisterUser(User *user, int &JumlahUser)
{
    try
    {
        if (JumlahUser >= 2)
            throw string("User penuh");

        cout << "\n========================\n";
        cout << "      REGISTER USER\n";
        cout << "========================\n";
```

```
        cout << "Masukkan nama baru: ";
        getline(cin, (user + JumlahUser)->username);
        if((user + JumlahUser)->username == "")
        throw string("Username tidak boleh kosongan");

        cout << "Masukkan password baru: ";
        getline(cin, (user + JumlahUser)->password);
        if((user + JumlahUser)->password == "")
        throw string("Password tidak boleh kosongan");

        (user + JumlahUser)->role = "user";
        JumlahUser++;

        cout << "Registrasi berhasil! Silakan login\n";
    }
    catch (string &e)
    {
        cout << "Error: " << e << endl;
    }
}

bool LoginUser(User *user, int &JumlahUser, string &InputNama, string
&role_login)
{
    string InputPassword;
    int percobaan_login = 0;

    while (percobaan_login < 3)
    {
        try
        {
            cout << "\n===================\n";
            cout << "        LOGIN\n";
            cout << "===================\n";

            cout << "Masukkan username: ";
            getline(cin, InputNama);
            if (InputNama == "")
            throw string("Username tidak boleh kosong");

            cout << "Masukkan password: ";
```

```
            getline(cin, InputPassword);
            if (InputPassword == "")
            throw string("Password tidak boleh kosong");

            for (int i = 0; i < JumlahUser; i++)
            {
                if (InputNama == (user + i)->username &&
InputPassword == (user + i)->password)
                {
                    cout << "Berhasil login! Selamat datang " <<
InputNama << endl;
                    role_login = (user + i)->role;
                    return true;
                }
            }

            throw string("Username atau password salah");
        }
        catch (string &e)
        {
            percobaan_login++;
            cout << "Error: " << e << endl;
            cout << "Sisa percobaan: " << (3 - percobaan_login) <<
endl;
        }
    }

    cout << "Gagal 3x! Akses ditolak!\n";
    return false;
}

void TambahDataKamera(Kamera *data, int &JumlahData)
{
    try
    {
        if (JumlahData == 10)
            throw string("Data kamera penuh!");

        cout <<
"\n=================================================================
======================" << endl;
        cout << "                          TAMBAH DATA KAMERA
```

```
TWISTERS CAM                                    " << endl;
        cout <<
"=================================================================
====================" << endl;

        cout << "ID Kamera: ";
        cin >> (data + JumlahData)->ID_Kamera;

        if ((data + JumlahData)->ID_Kamera <= 0)
            throw string("ID tidak valid");

        cin.ignore();

        cout << "Nama Kamera: ";
        getline(cin, (data + JumlahData)->NamaKamera);

        cout << "Harga sewa/hari: Rp";
        cin >> (data + JumlahData)->HargaSewa;

        if ((data + JumlahData)->HargaSewa < 0)
            throw string("Harga tidak valid");

        cin.ignore();

        (data + JumlahData)->status.kondisi = "Tersedia";
        JumlahData++;

        cout << "Data berhasil ditambahkan\n";
    }
    catch (string &e)
    {
        cout << "Error: " << e << endl;
    }
}

void TampilkanDataKamera(Kamera *data, int &JumlahData)
{
    if (JumlahData == 0)
    {
        cout << "Data kamera belum tersedia";
        return;
    }
```

```
    cout <<
"\n=================================================================
=====================\n";
    cout << "                                DATA KAMERA TWISTERS CAM\n";
    cout <<
"=================================================================
====================\n";
    cout << left << setw(5) << "No";
    cout << setw(10) << "ID";
    cout << setw(20) << "Nama Kamera";
    cout << setw(20) << "Harga Sewa/hari";
    cout << setw(12) << "Status";
    cout << endl;
    cout <<
"-----------------------------------------------------------------
------------------\n";

    for (int i = 0; i < JumlahData; i++)
    {
        cout << left << setw(5) << i + 1;
        cout << setw(10) << (data + i)->ID_Kamera;
        cout << setw(20) << (data + i)->NamaKamera;
        cout << "Rp" << setw(17) << (data + i)->HargaSewa;
        cout << setw(12) << (data + i)->status.kondisi << endl;
    }
    cout <<
"=================================================================
====================\n";
}

void UbahDataKamera(Kamera *data, int &JumlahData)
{
    try
    {
        if (JumlahData == 0)
            throw string("Data kosong");

        TampilkanDataKamera(data, JumlahData);

        int index;
        cout << "Pilih nomor kamera: ";
        cin >> index;
```

```
        index--;

        if (index < 0 || index >= JumlahData)
            throw string("Index tidak valid");

        cout << "ID baru: ";
        cin >> (data + index)->ID_Kamera;

        cin.ignore();

        cout << "Nama baru: ";
        getline(cin, (data + index)->NamaKamera);

        cout << "Harga baru: ";
        cin >> (data + index)->HargaSewa;

        cin.ignore();

        cout << "Status: ";
        getline(cin, (data + index)->status.kondisi);

        cout << "Berhasil diubah\n";
    }
    catch (string &e)
    {
        cout << "Error: " << e << endl;
    }
}

void HapusDataKamera(Kamera *data, int &JumlahData)
{
    try
    {
        if (JumlahData == 0)
            throw string("Data kosong");

        TampilkanDataKamera(data, JumlahData);

        int index;
        char konfir;

        cout << "Pilih nomor: ";
```

```
        cin >> index;
        index--;

        if (index < 0 || index >= JumlahData)
            throw string("Index tidak valid");

        cout << "Yakin? (Y/N): ";
        cin >> konfir;

        if (konfir != 'Y' && konfir != 'y')
            throw string("Dibatalkan");

        for (int i = index; i < JumlahData - 1; i++)
        {
            *(data + i) = *(data + i + 1);
        }

        JumlahData--;

        cout << "Data dihapus\n";
    }
    catch (string &e)
    {
        cout << "Error: " << e << endl;
    }
}

int KembalikanKamera(int harga, int hari)
{
    try
    {
        if (hari < 0)
            throw string("Hari tidak valid");

        if (hari == 0)
            return 0;

        return harga + KembalikanKamera(harga, hari - 1);
    }
    catch (string &e)
    {
        cout << "Error: " << e << endl;
```

```
        return 0;
    }
}

int KembalikanKamera(int harga, int lamaHari, float diskonSewa)
{
    return (harga * lamaHari) - ((harga * lamaHari) * diskonSewa);
}

void BubbleSortKameraDescending(Kamera *data, int &JumlahData)
{
    for (int i = 0; i < JumlahData - 1; i++)
    {
        for (int j = 0; j < JumlahData - i - 1; j++)
        {
            if ((data + j)->NamaKamera < (data + j + 1)->NamaKamera)
            {
                Kamera temp = *(data + j);
                *(data + j) = *(data + j + 1);
                *(data + j + 1) = temp;
            }
        }
    }
}

void SelectionSortHargaAscending(Kamera *data, int &JumlahData)
{
    for (int i = 0; i < JumlahData - 1; i++)
    {
        int min = i;
        for (int j = i + 1; j < JumlahData; j++)
        {
            if ((data + j)->HargaSewa < (data + min)->HargaSewa)
                min = j;
        }
        Kamera temp = *(data + i);
        *(data + i) = *(data + min);
        *(data + min) = temp;
    }
}

int PartitionID(Kamera *data, int low, int high)
```

```
{
    int pivot = (data + high)->ID_Kamera;
    int i = low - 1;

    for (int j = low; j < high; j++)
    {
        if ((data + j)->ID_Kamera < pivot)
        {
            i++;
            Kamera temp = *(data + i);
            *(data + i) = *(data + j);
            *(data + j) = temp;
        }
    }

    Kamera temp = *(data + i + 1);
    *(data + i + 1) = *(data + high);
    *(data + high) = temp;

    return i + 1;
}

void QuickSortID(Kamera *data, int low, int high)
{
    if (low < high)
    {
        int p = PartitionID(data, low, high);
        QuickSortID(data, low, p - 1);
        QuickSortID(data, p + 1, high);
    }
}

int LinearSearchKamera(Kamera *data, int JumlahData, string key)
{
    for (int i = 0; i < JumlahData; i++)
    {
        if ((data + i)->NamaKamera == key)
            return i;
    }
    return -1;
}
```

```
int BinarySearchID(Kamera *data, int JumlahData, int key)
{
    int kiri = 0, kanan = JumlahData - 1;

    while (kiri <= kanan)
    {
        int tengah = (kiri + kanan) / 2;

        if ((data + tengah)->ID_Kamera == key)
            return tengah;
        else if ((data + tengah)->ID_Kamera < key)
            kiri = tengah + 1;
        else
            kanan = tengah - 1;
    }

    return -1;
}

int main()
{
    Kamera data[10];
    User user[2];
    int JumlahUser = 0, JumlahData = 0;

    user[JumlahUser++] = {"LARRY", "068", "admin"};

    bool MenuAwal = true;

    while (MenuAwal)
    {
        int PilihMenu;
        string InputNama, role_login;

        cout << "\n==========================\n";
        cout << "  HOME PAGE TWISTERS CAM\n";
        cout << "==========================\n";
        cout << "1. Register\n2. Login\n3. Keluar\n";
        cout << "Pilih (1-3): ";

        try
        {
```

```
            if (!(cin >> PilihMenu))
                throw string("Input harus angka!");

            cin.ignore(10000, '\n');

            switch (PilihMenu)
            {
            case 1:
                RegisterUser(user, JumlahUser);
                break;

            case 2:
            {
                if (!LoginUser(user, JumlahUser, InputNama,
role_login))
                    throw string("Login gagal 3x");

                if (role_login == "admin")
                {
                    int pilihanAdmin;

                    do
                    {
                        cout << "\n======================\n";
                        cout << "        MENU ADMIN\n";
                        cout << "======================\n";
                        cout << "1. Tambah Data Kamera\n";
                        cout << "2. Tampilkan Data Kamera\n";
                        cout << "3. Ubah Data Kamera\n";
                        cout << "4. Hapus Data Kamera\n";
                        cout << "5. Sort Nama Kamera\n";
                        cout << "6. Sort Harga Kamera\n";
                        cout << "7. Sort ID Kamera\n";
                        cout << "8. Search Nama Kamera\n";
                        cout << "9. Search ID Kamera\n";
                        cout << "10. Logout\n";
                        cout << "Pilih (1-10): ";

                        if (!(cin >> pilihanAdmin))
                            throw string("Input admin harus angka!");

                        cin.ignore(10000, '\n');
```

```
                        switch (pilihanAdmin)
                        {
                        case 1:
                            TambahDataKamera(data, JumlahData);
                            break;

                        case 2:
                            TampilkanDataKamera(data, JumlahData);
                            break;

                        case 3:
                            UbahDataKamera(data, JumlahData);
                            break;

                        case 4:
                            HapusDataKamera(data, JumlahData);
                            break;

                        case 5:
                            BubbleSortKameraDescending(data,
JumlahData);
                            TampilkanDataKamera(data, JumlahData);
                            break;

                        case 6:
                            SelectionSortHargaAscending(data,
JumlahData);
                            TampilkanDataKamera(data, JumlahData);
                            break;

                        case 7:
                            QuickSortID(data, 0, JumlahData - 1);
                            TampilkanDataKamera(data, JumlahData);
                            break;

                        case 8:
                        {
                            string cari;
                            cout << "Masukkan nama kamera: ";
                            getline(cin, cari);
```

```
                            int hasil = LinearSearchKamera(data,
JumlahData, cari);

                            if (hasil == -1)
                                throw string("Data tidak ditemukan");

                            cout << "ID: " << (data +
hasil)->ID_Kamera << endl;
                            cout << "Nama: " << (data +
hasil)->NamaKamera << endl;
                            cout << "Harga: Rp" << (data +
hasil)->HargaSewa << endl;
                            break;
                        }

                        case 9:
                        {
                            if (JumlahData == 0)
                                throw string("Data kosong");

                            QuickSortID(data, 0, JumlahData - 1);

                            int cariID;
                            cout << "Masukkan ID kamera: ";

                            if (!(cin >> cariID))
                                throw string("ID harus angka!");

                            int hasil = BinarySearchID(data,
JumlahData, cariID);

                            if (hasil == -1)
                                throw string("Data tidak ditemukan");

                            cout << "ID: " << (data +
hasil)->ID_Kamera << endl;
                            cout << "Nama: " << (data +
hasil)->NamaKamera << endl;
                            cout << "Harga: Rp" << (data +
hasil)->HargaSewa << endl;
                            break;
                        }
```

```
                    case 10:
                        cout << "Logout admin...\n";
                        break;

                    default:
                        cout << "Menu tidak valid\n";
                    }

                } while (pilihanAdmin != 10);
            }
            else
            {
                int pilihanUser;

                do
                {
                    cout << "\n======================\n";
                    cout << "        MENU USER\n";
                    cout << "======================\n";
                    cout << "1. Lihat Data Kamera\n";
                    cout << "2. Sewa Kamera\n";
                    cout << "3. Kembalikan Kamera\n";
                    cout << "4. Logout\n";
                    cout << "Pilih (1-4): ";

                    if (!(cin >> pilihanUser))
                        throw string("Input harus angka!");

                    cin.ignore(10000, '\n');

                    switch (pilihanUser)
                    {
                    case 1:
                        TampilkanDataKamera(data, JumlahData);
                        break;

                    case 2:
                    {
                        if (JumlahData == 0)
                            throw string("Data kosong");
```

```
                    TampilkanDataKamera(data, JumlahData);

                    int nomor;
                    cout << "Pilih kamera: ";

                    if (!(cin >> nomor))
                        throw string("Input harus angka");

                    nomor--;

                    if (nomor < 0 || nomor >= JumlahData)
                        throw string("Index tidak valid");

                    if ((data + nomor)->status.kondisi !=
"Tersedia")
                    {
                        cout << "Kamera sudah disewa\n";
                        break;
                    }

                    cout << "Lama sewa: ";
                    cin >> (data + nomor)->LamaSewa;

                    (data + nomor)->status.kondisi =
"Disewa";
                    (data + nomor)->NamaPenyewa = InputNama;

                    cout << "Berhasil disewa\n";
                    break;
                }

                case 3:
                {
                    if (JumlahData == 0)
                        throw string("Data kosong");

                    TampilkanDataKamera(data, JumlahData);

                    int nomor;
                    cout << "Pilih kamera: ";

                    if (!(cin >> nomor))
```

```
                                    throw string("Input harus angka");

                                nomor--;

                                if (nomor < 0 || nomor >= JumlahData)
                                    throw string("Index tidak valid");

                                if ((data + nomor)->NamaPenyewa !=
InputNama)
                                    throw string("Bukan kamera milik
anda");

                                int total = KembalikanKamera((data +
nomor)->HargaSewa, (data + nomor)->LamaSewa);

                                if (total >= 750000)
                                {
                                    total = KembalikanKamera((data +
nomor)->HargaSewa, (data + nomor)->LamaSewa, 0.15);
                                }

                                cout << "Total bayar: Rp" << total <<
endl;

                                (data + nomor)->status.kondisi =
"Tersedia";
                                (data + nomor)->NamaPenyewa = "";
                                (data + nomor)->LamaSewa = 0;

                                cout << "Berhasil dikembalikan\n";
                                break;
                            }

                            case 4:
                                cout << "Logout user...\n";
                                break;

                            default:
                                cout << "Menu tidak valid\n";
                            }

                        } while (pilihanUser != 4);
```

```
                }
                break;
            }

            case 3:
                cout << "Program selesai\n";
                MenuAwal = false;
                break;

            default:
                cout << "Menu utama tidak valid\n";
            }
        }
        catch (string &e)
        {
            cout << "\n[ERROR] " << e << "\n";
            cin.clear();
            cin.ignore(10000, '\n');
        }
    }

    return 0;
}
```

4. Hasil Output

```
============================
  HOME PAGE TWISTERS CAM
============================
1. Register
2. Login
3. Keluar
Pilih (1-3): 2

====================
       LOGIN
====================
Masukkan username: LARRY
Masukkan password: 068
Berhasil login! Selamat datang LARRY
```

Admin melakukan login

```
Pilih (1-10): 1

==========================================================================
                        TAMBAH DATA KAMERA TWISTERS CAM
==========================================================================
ID Kamera: 10
Nama Kamera: FUJIFILM 95
Harga sewa/hari: Rp225000
Data berhasil ditambahkan
```

Admin menambah data kamera

```
Pilih (1-10): 2

==========================================================================
                         DATA KAMERA TWISTERS CAM
==========================================================================
No   ID        Nama Kamera         Harga Sewa/hari     Status
--------------------------------------------------------------------------
1    1         CANON ZOOM          Rp576000            Tersedia
2    2         SONY A7             Rp500000            Tersedia
3    3         SONY XS             Rp340000            Tersedia
4    4         FUJIFILM 74         Rp155000            Tersedia
5    5         FUJIFILM 89         Rp200000            Tersedia
6    10        FUJIFILM 95         Rp225000            Tersedia
7    9         NIKON SUPER         Rp400000            Tersedia
8    8         NIKON GAMMA         Rp325000            Tersedia
9    7         CANON 1200D         Rp350000            Tersedia
10   6         CANON 600D          Rp300000            Tersedia
==========================================================================
```

Admin melihat data kamera

```
Pilih (1-10): 3

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera          Harga Sewa/hari      Status
------------------------------------------------------------------------------------------
1     1         CANON ZOOM           Rp576000            Tersedia
2     2         SONY A7              Rp500000            Tersedia
3     3         SONY XS              Rp340000            Tersedia
4     4         FUJIFILM 74          Rp155000            Tersedia
5     5         FUJIFILM 89          Rp200000            Tersedia
6     10        FUJIFILM 95          Rp225000            Tersedia
7     9         NIKON SUPER          Rp400000            Tersedia
8     8         NIKON GAMMA          Rp325000            Tersedia
9     7         CANON 1200D          Rp350000            Tersedia
10    6         CANON 600D           Rp300000            Tersedia
==========================================================================================
Pilih nomor kamera yang ingin diubah: 10
ID Kamera baru: 11
Nama Kamera baru: CANON X
Harga sewa/hari: Rp475000
Status (Tersedia/Disewa): Tersedia
Data kamera berhasil diubah!

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera          Harga Sewa/hari      Status
------------------------------------------------------------------------------------------
1     1         CANON ZOOM           Rp576000            Tersedia
2     2         SONY A7              Rp500000            Tersedia
3     3         SONY XS              Rp340000            Tersedia
4     4         FUJIFILM 74          Rp155000            Tersedia
5     5         FUJIFILM 89          Rp200000            Tersedia
6     10        FUJIFILM 95          Rp225000            Tersedia
7     9         NIKON SUPER          Rp400000            Tersedia
8     8         NIKON GAMMA          Rp325000            Tersedia
9     7         CANON 1200D          Rp350000            Tersedia
10    11        CANON X              Rp475000            Tersedia
==========================================================================================
```

Admin mengubah data kamera

```
Pilih (1-10): 4

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     1         CANON ZOOM          Rp576000            Tersedia
2     2         SONY A7             Rp500000            Tersedia
3     3         SONY XS             Rp340000            Tersedia
4     4         FUJIFILM 74         Rp155000            Tersedia
5     5         FUJIFILM 89         Rp200000            Tersedia
6     10        FUJIFILM 95         Rp225000            Tersedia
7     9         NIKON SUPER         Rp400000            Tersedia
8     8         NIKON GAMMA         Rp325000            Tersedia
9     7         CANON 1200D         Rp350000            Tersedia
10    11        CANON X             Rp475000            Tersedia
==========================================================================================
Pilih nomor kamera yang ingin dihapus: 5
Yakin ingin hapus data kamera ini? (Y/N): y
Data kamera berhasil dihapus!

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     1         CANON ZOOM          Rp576000            Tersedia
2     2         SONY A7             Rp500000            Tersedia
3     3         SONY XS             Rp340000            Tersedia
4     4         FUJIFILM 74         Rp155000            Tersedia
5     10        FUJIFILM 95         Rp225000            Tersedia
6     9         NIKON SUPER         Rp400000            Tersedia
7     8         NIKON GAMMA         Rp325000            Tersedia
8     7         CANON 1200D         Rp350000            Tersedia
9     11        CANON X             Rp475000            Tersedia
==========================================================================================
```

Admin menghapus data kamera

```
Pilih (1-10): 5
Sort Nama Kamera berhasil

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     3         SONY XS             Rp340000            Tersedia
2     2         SONY A7             Rp500000            Tersedia
3     9         NIKON SUPER         Rp400000            Tersedia
4     8         NIKON GAMMA         Rp325000            Tersedia
5     10        FUJIFILM 95         Rp225000            Tersedia
6     4         FUJIFILM 74         Rp155000            Tersedia
7     1         CANON ZOOM          Rp576000            Tersedia
8     11        CANON X             Rp475000            Tersedia
9     7         CANON 1200D         Rp350000            Tersedia
==========================================================================================
```

Sort nama kamera secara descending

```
Pilih (1-10): 6
Sort Harga Kamera berhasil

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     4         FUJIFILM 74         Rp155000            Tersedia
2     10        FUJIFILM 95         Rp225000            Tersedia
3     8         NIKON GAMMA         Rp325000            Tersedia
4     3         SONY XS             Rp340000            Tersedia
5     7         CANON 1200D         Rp350000            Tersedia
6     9         NIKON SUPER         Rp400000            Tersedia
7     11        CANON X             Rp475000            Tersedia
8     2         SONY A7             Rp500000            Tersedia
9     1         CANON ZOOM          Rp576000            Tersedia
==========================================================================================
```

Sort harga sewa kamera secara ascending

```
Pilih (1-10): 7
Sort ID Kamera berhasil

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     1         CANON ZOOM          Rp576000            Tersedia
2     2         SONY A7             Rp500000            Tersedia
3     3         SONY XS             Rp340000            Tersedia
4     4         FUJIFILM 74         Rp155000            Tersedia
5     7         CANON 1200D         Rp350000            Tersedia
6     8         NIKON GAMMA         Rp325000            Tersedia
7     9         NIKON SUPER         Rp400000            Tersedia
8     10        FUJIFILM 95         Rp225000            Tersedia
9     11        CANON X             Rp475000            Tersedia
==========================================================================================
```

Sort ID secara ascending

```
Pilih (1-10): 8
Masukkan nama kamera: FUJIFILM 74
Data kamera ditemukan
ID Kamera: 4
Nama Kamera: FUJIFILM 74
Harga sewa/hari: Rp155000
Status: Tersedia
```

Search nama kamera

```
Pilih (1-10): 9
Masukkan ID Kamera:
4
Data kamera ditemukan
ID Kamera: 4
Nama Kamera: FUJIFILM 74
Harga sewa/hari: Rp155000
Status: Tersedia
```

Search ID kamera

```
============================
  HOME PAGE TWISTERS CAM
============================
1. Register
2. Login
3. Keluar
Pilih (1-3): 1

========================
      REGISTER USER
========================
Masukkan nama baru: Muhammad Noval Arifinnur
Masukkan password baru: Opang
Registrasi berhasil! Silakan login
```

Register User

```
============================
  HOME PAGE TWISTERS CAM
============================
1. Register
2. Login
3. Keluar
Pilih (1-3): 2

====================
       LOGIN
====================
Masukkan username: Muhammad Noval Arifinnur
Masukkan password: Opang
Berhasil login! Selamat datang Muhammad Noval Arifinnur
```

Login User yang sudah register

```
=======================
        MENU USER
=======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 1

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     1         CANON ZOOM          Rp576000            Tersedia
2     2         SONY A7             Rp500000            Tersedia
3     3         SONY XS             Rp340000            Tersedia
4     4         FUJIFILM 74         Rp155000            Tersedia
5     7         CANON 1200D         Rp350000            Tersedia
6     8         NIKON GAMMA         Rp325000            Tersedia
7     9         NIKON SUPER         Rp400000            Tersedia
8     10        FUJIFILM 95         Rp225000            Tersedia
9     11        CANON X             Rp475000            Tersedia
==========================================================================================
```

User melihat data kamera yang ada

```
=======================
        MENU USER
=======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 2

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     1         CANON ZOOM          Rp576000            Tersedia
2     2         SONY A7             Rp500000            Tersedia
3     3         SONY XS             Rp340000            Tersedia
4     4         FUJIFILM 74         Rp155000            Tersedia
5     7         CANON 1200D         Rp350000            Tersedia
6     8         NIKON GAMMA         Rp325000            Tersedia
7     9         NIKON SUPER         Rp400000            Tersedia
8     10        FUJIFILM 95         Rp225000            Tersedia
9     11        CANON X             Rp475000            Tersedia
==========================================================================================
Pilih nomor kamera yang ingin disewa: 5
Sewa berapa hari: 3
Kamera berhasil disewa!
```

User menyewa kamera

```
=====================
       MENU USER
=====================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 3

=================================================================================
                            DATA KAMERA TWISTERS CAM
=================================================================================
No   ID        Nama Kamera         Harga Sewa/hari     Status
---------------------------------------------------------------------------------
1    1         CANON ZOOM          Rp576000            Disewa
2    2         SONY A7             Rp500000            Tersedia
3    3         SONY XS             Rp340000            Tersedia
4    4         FUJIFILM 74         Rp155000            Tersedia
5    7         CANON 1200D         Rp350000            Tersedia
6    8         NIKON GAMMA         Rp325000            Tersedia
7    9         NIKON SUPER         Rp400000            Tersedia
8    10        FUJIFILM 95         Rp225000            Tersedia
9    11        CANON X             Rp475000            Tersedia
=================================================================================
Pilih nomor kamera yang Anda sewa untuk dikembalikan: 1
Total harga sewa yang harus dibayar: Rp1958400
Kamera berhasil dikembalikan
```

User mengembalikan kamera yang disewa

```
=======================
       MENU USER
=======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 3

[ERROR] Data kosong
```

```
=======================
       MENU USER
=======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 2

[ERROR] Data kosong
```

```
=========================
        MENU ADMIN
=========================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Sort Nama Kamera
6. Sort Harga Kamera
7. Sort ID Kamera
8. Search Nama Kamera
9. Search ID Kamera
10. Logout
Pilih (1-10): 3
Error: Data kosong
```

```
=============================
  HOME PAGE TWISTERS CAM
=============================
1. Register
2. Login
3. Keluar
Pilih (1-3): 1
Error: User penuh
```

```
==========================================================================
                    TAMBAH DATA KAMERA TWISTERS CAM
==========================================================================
ID Kamera: SATU
Error: ID tidak valid
```

```
====================
       LOGIN
====================
Masukkan username: LARRY
Masukkan password:
Error: Password tidak boleh kosong
Sisa percobaan: 1
```

```
====================
       LOGIN
====================
Masukkan username:
Error: Username tidak boleh kosong
Sisa percobaan: 2
```

```
========================
      REGISTER USER
========================
Masukkan nama baru: OWEN
Masukkan password baru:
Error: Password tidak boleh kosongan
```

```
========================
      REGISTER USER
========================
Masukkan nama baru:
Error: Username tidak boleh kosongan
```

5. Langkah-langkah GIT

5.1 GIT Add

```
PS C:\praktikum-apl> git add "post-test/post-test-apl-7/2509106068-NATHANAEL LARRY GLASIA-PT-7.cpp"
PS C:\praktikum-apl>  git add "post-test/post-test-apl-7/2509106068-NATHANAEL LARRY GLASIA-PT-7.exe"
```

Menambahkan file ke staging area untuk di commit

5.2 GIT Commit

```
PS C:\praktikum-apl> git commit -m "error handling pt 7 smg fix"
[main f30f174] error handling pt 7 smg fix
 2 files changed, 654 insertions(+)
 create mode 100644 post-test/post-test-apl-7/2509106068-NATHANAEL LARRY GLASIA-PT-7.cpp
 create mode 100644 post-test/post-test-apl-7/2509106068-NATHANAEL LARRY GLASIA-PT-7.exe
```

Menyimpan perubahan yang ada di staging area ke repositori lokal dengan pesan

5.3 Git Push

```
PS C:\praktikum-apl> git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 4 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 96.40 KiB | 1.34 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/larrynathanx/praktikum-apl.git
   abba7a2..f30f174  main -> main
```

Mengunggah perubahan dari repositori lokal ke repositori jarak jauh (GitHub)